

Draft - expert review pending

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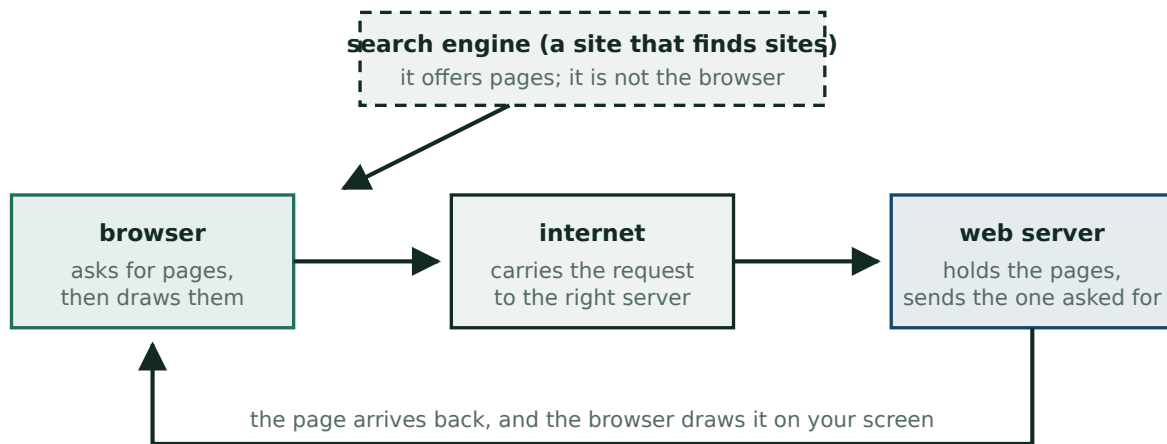
Topic 6.1 - Browsers, search engines, and online communication

A real classroom situation

A trainee teacher in Khairpur needs a lesson idea on photosynthesis for Class 6. She opens her phone, types "photosynthesis" into the browser's address bar, taps the first result - an advertisement - and concludes "the internet has nothing good on this." Her supervisor, watching, takes the same phone and in four minutes has a university worksheet, a labelled diagram and a five-minute video. Same internet, same phone, same word. The difference was never the device: it was knowing that the browser and the search engine are two different tools, that the address bar is not a search box, and that the first result is a bid, not a recommendation. This topic is those four minutes.

The journey of a web request

The browser carries you; the search engine is itself a site that finds other sites



Same internet, same phone, same word: the difference is knowing which tool does which job.

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Explanation

Web browsers

A web browser is the application that asks for, receives and displays web pages: Chrome, Firefox, Safari, Edge and their relatives. When you type an address (a URL) and press enter, the browser sends the request across the internet, the server holding that page sends it back, and the browser draws it on your screen - the journey in the figure. The browser also keeps your history, your bookmarks and your logins, which is why Unit 4 treated the browser as a security boundary: its padlock and its warnings are the first checkpoint of safe browsing.

A search engine - Google, Bing, DuckDuckGo and others - is not the browser and not the internet. It is itself a website: one that has read enormous parts of the web in advance and keeps an index, so that when you type words it can offer pages that match. The browser carries you; the search engine finds. Because a search engine ranks by a mixture of relevance, popularity and - importantly - payment, the first result is a bid as often as a judgement. The professional habit is to read the first five results' titles and addresses, not to tap the first.

Searching well is a teachable skill. Put the specific words in ("photosynthesis Class 6 worksheet" beats "plants"); use the minus sign to drop noise ("jaguar - car"); use quotation marks for exact phrases; and prefer site-limited searches ("site:edu.pk" or "site:gov.pk") when you want Pakistani institutional material. Then judge each result the way Topic 6.2 teaches: who made this page, and why.

Communication tools: match, do not just master

Tool	What it is best for	One classroom use
email	formal, record-keeping communication	sending the term report to parents
instant messaging	quick questions inside a group	class group announcements
video meeting	live teaching and meetings across distance	remote parent meeting
discussion forum	extended written discussion, every voice heard	weekly homework questions
learning platform	structured courses with materials and tracking	the school's assignment portal

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Online communication tools

The internet's second everyday job is carrying communication, and its tools divide by what each is good at. **Email** is formal and keeps records: anything a school might need to prove later belongs there. **Instant messaging and chat** (WhatsApp and its relatives) is fast and informal - excellent for announcements and quick questions, poor for records and terrible for anything confidential, as Unit 4's forwarding stories showed. **Video meetings** carry live teaching and meetings across distance, at the cost of bandwidth, which in much of Sindh means audio-only in practice. **Discussion forums** support extended written discussion where every voice can be heard, not just the loudest. And **learning platforms** bundle materials, assignments and tracking into one place - the school's structured online home.

The skill is matching, not mastering: choosing the tool that fits the purpose, the audience and the record the school needs. The parent who needs a formal record gets an email; the colleague who needs an answer in five minutes gets a chat message; the class that needs to discuss gets a forum, not a shouty group.

Activity: The four-minute search

Work in pairs. About 20 minutes.

1. Your teacher gives each pair the same need: "a Class 6 lesson idea on photosynthesis, in English, usable in a school with no lab."
2. Pair A searches as the trainee in the story did: one word, first result. Pair B plans first: specific words, a site limit, and a plan to read five titles before tapping.
3. Both pairs have four minutes. Note what each finds and how usable it is for the stated need.
4. Compare, then swap roles with a new need ("past papers for Class 8 science, Sindh board"). End by writing three search rules you would teach a Class 7 pupil.

Check your understanding

1. Define a web browser and a search engine in one sentence each, and state the key difference. (*Remember*)
2. Explain why the first search result is "a bid as often as a judgement". (*Understand*)
3. A school must send formal term reports to parents and give pupils a place to discuss homework questions. Match each need to the right tool and justify each choice. (*Apply*)
4. Write a better search for: lesson material on the water cycle for Class 5, from Pakistani educational institutions only. (*Apply*)

Summary

The browser asks for, receives and draws web pages; the search engine is itself a site that keeps an index of the web and offers matches - and its first result mixes relevance with payment, so read before you tap. Searching deliberately means specific words, minus signs, quoted phrases and site limits.

Communication tools divide by purpose: email for the formal record, chat for speed, video for live distance work, forums for extended discussion, platforms for structured courses. The skill is matching the tool to the purpose.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can explain the difference between a browser and a search engine.
- ☐ I can plan a search with specific words and a site limit, and read five titles before tapping.
- ☐ I can match a communication need to the right tool with a reason.

Try this at your practicum school

Find out which communication tools the school actually uses for parents (email, SMS, chat groups, the diary). Note which purposes each serves well and which poorly, and one change you would recommend.

Summative task

Prepare a one-page "search and communicate" guide for a colleague: five search rules with a worked example (a real need, the planned search, and why the chosen results are trustworthy), and a tool-matching table for the school's five common communication needs.

Marking guide (10 marks):

- Five search rules, each workable and distinct: 3 marks.
- The worked example shows a planned search and judged results, not a first tap (Analyse): 4 marks.
- The tool-matching table fits purposes with reasons: 2 marks.

- Fits one page: 1 mark.

Further reading

- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 5. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- Khan Academy. *Computers and the Internet* (the internet units). <https://www.khanacademy.org/computing/computers-and-the-internet>
- National Cybersecurity Alliance. *Stay Safe Online*. <https://staysafeonline.org/>

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